

Personal Project Dataset Validation Report

Minor AI for Society

Casey Stijlaart

Table of Contents

1. Introduction	1
2. Purpose	1
3. Used Datasets	1
4. Validation Workflow	2
5. Validation Criteria	2
6. Dataset 1: Indoor Plant Health and Growth Factors	3
6.1. Findings	3
6.2. Key Metrics	3
6.3. Proof Images	3
7. Dataset 2: Agriculture Dataset with Target	5
7.1. Findings	5
7.2. Key Metrics	5
7.3. Proof Images	5
8. Dataset 3: Plant Health Biosensor Dataset	6
8.1. Findings	6
8.2. Key Metrics	6
8.3. Proof Images	6
9. Dataset 4: Plant Health Data	7
9.1. Findings	7
9.2. Key Metrics	7
9.3. Engineered Version	7
9.4. Proof Images	7
10. Cross-Dataset Comparison (TinyML / TinyRegression / Anomaly Detection)	10
11. Executive Summary	11
12. Final Recommendation	11
13. References	12

1. Introduction

Public datasets enable rapid experimentation with machine learning, but their suitability for real-world and constrained applications (e.g., TinyML) varies significantly. For this Minor AI for Society project, dataset validation is therefore a critical step to ensure reliable and deployable model performance.

This report evaluates four plant-related datasets to determine their suitability for lightweight machine learning models. Each dataset is assessed using a consistent workflow that combines data quality checks with practical model testing, including baseline models, compact neural networks (TinyMLP), regression variants, and anomaly detection.

The goal is to identify which dataset best supports accurate, efficient, and realistic deployment within constrained environments.

2. Purpose

This document summarizes the validation work for the four collected datasets and evaluates whether each dataset is suitable for the personal project goals.

The validation focuses on:

- feature availability for the target use case,
- basic data quality checks,
- baseline model behavior,
- compatibility with small models (TinyMLP / TinyRegression), and
- anomaly-detection potential.

3. Used Datasets

Kaggle offers a variety of datasets regarding to the given objective. While many had potential, not all were suitable. In order to find suitable datasets, it was of great importance to have certain key features be included, such as plant health, soil moisture, temperature and humidity.

The validation in this report uses the following public datasets:

1. Indoor Plant Health and Growth Dataset (Kaggle): <https://www.kaggle.com/datasets/souvikrana17/indoor-plant-health-and-growth-dataset>
2. Edge Assisted Agricultural Sensor Dataset (Kaggle): <https://www.kaggle.com/datasets/colabsss/edge-assisted-agricultural-sensor-dataset>
3. Plant Health Biosensor Dataset (Kaggle): <https://www.kaggle.com/datasets/programmer3/plant-health-biosensor-dataset>
4. Plant Health Diseases Dataset (Kaggle): <https://www.kaggle.com/datasets/muqaddasejaz/plant-health-diseases-dataset>

4. Validation Workflow

The following steps were used for each dataset to keep the validation process consistent and reproducible:

1. Imported the raw dataset and inspected schema/column names.
2. Checked required fields for the project use case (sensor-like inputs + plant health target).
3. Reviewed data quality (missing values, duplicates, data types, and target distribution).
4. Performed basic feature preparation and train/test splitting.
5. Trained a baseline `RandomForestClassifier` to estimate practical signal strength.
6. Trained a compact `TinyMLP` model to evaluate TinyML feasibility.
7. Where applicable, trained a compact regression variant (`TinyRegression`) and recorded `MSE`, `MAE`, and `R2`.
8. Ran anomaly detection (`IsolationForest`) to evaluate anomaly-detection suitability.
9. Compared the outcomes across datasets to rank suitability for deployment constraints.

This workflow was applied to provide both data-quality validation and model-readiness validation before selecting a final dataset.

5. Validation Criteria

Each dataset was reviewed using the same checklist:

- Required fields available (light, soil moisture, humidity, ambient temperature, plant health or equivalent target).
- Missing values and duplicates.
- Data types and label format.
- Class balance / target signal quality.
- Baseline Random Forest test performance.
- TinyMLP / TinyRegression feasibility.

6. Dataset 1: Indoor Plant Health and Growth Factors

6.1. Findings

- Baseline performance is low and unstable.
- Adding `Plant_ID` / plant-specific information produces only marginal improvement.
- Correlation analysis indicates weak relationships between input features and the health target.
- Feature importances are relatively flat, suggesting low signal strength.
- Adding extra columns (for example light exposure, soil type, pest presence) did not improve results and in some runs reduced performance.

6.2. Key Metrics

- Random Forest test accuracy: `0.2140`
- TinyMLP test accuracy: `0.2040`
- Anomaly detections: `132`

6.3. Proof Images

```
Dataset Validation Summary
-----
Rows used: 1000
Features used: ['Soil_Moisture', 'Humidity', 'Ambient_Temperature']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.3600
```

Figure 1. Indoor Plant Health Summary

```
Dataset Validation Summary
-----
Rows used: 1000
Features used: ['Soil_Moisture', 'Humidity', 'Ambient_Temperature', 'Plant_ID']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.3950
```

Figure 2. Indoor Plant Health With ID Summary

Train set report:				
	precision	recall	f1-score	support
Healthy	1.00	1.00	1.00	170
High Stress	1.00	1.00	1.00	328
Moderate Stress	1.00	1.00	1.00	302
accuracy			1.00	800
macro avg	1.00	1.00	1.00	800
weighted avg	1.00	1.00	1.00	800
Test set report:				
	precision	recall	f1-score	support
Healthy	0.40	0.14	0.21	43
High Stress	0.35	0.43	0.38	82
Moderate Stress	0.36	0.41	0.39	75
accuracy			0.36	200
macro avg	0.37	0.33	0.33	200
weighted avg	0.37	0.36	0.35	200

Figure 3. Indoor Plant Health Report

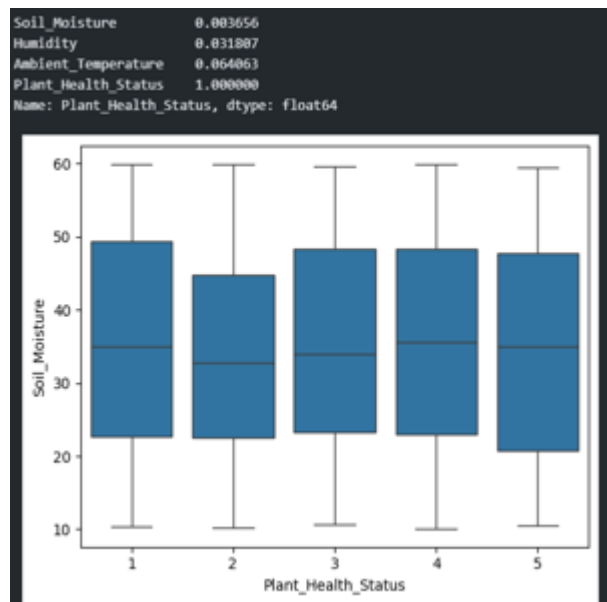


Figure 4. Indoor Plant Health Correlation

7. Dataset 2: Agriculture Dataset with Target

7.1. Findings

- The dataset can be used for broad experimentation, but predictive strength is limited.
- Baseline classification quality is low, indicating noisy or weakly informative feature-target mapping.
- TinyRegression results also indicate weak explanatory power ($R^2 < 0$).

7.2. Key Metrics

- Random Forest test accuracy: 0.3125
- TinyMLP test accuracy: 0.3640
- TinyRegression: MSE 0.7001, MAE 0.7157, R^2 -0.0730
- Anomaly detections: 218

7.3. Proof Images

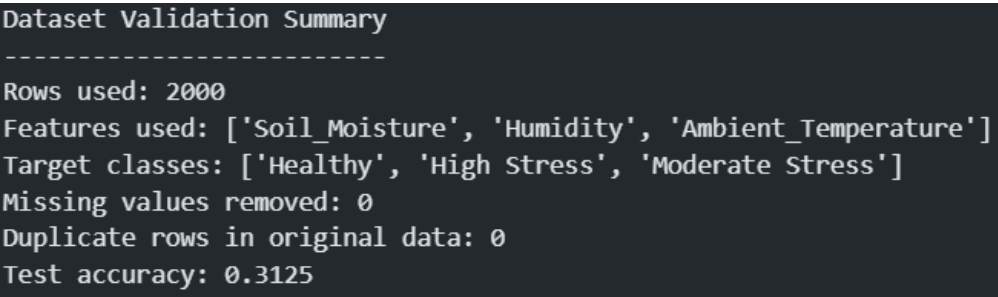


Figure 5. Agriculture Dataset Summary

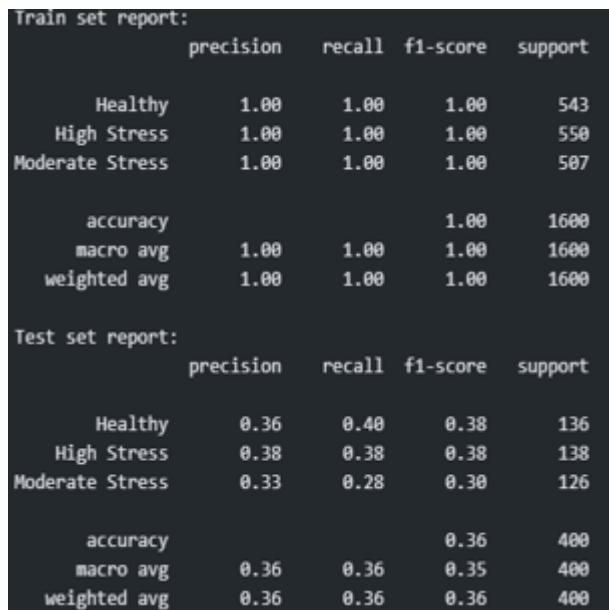


Figure 6. Agriculture Dataset Report

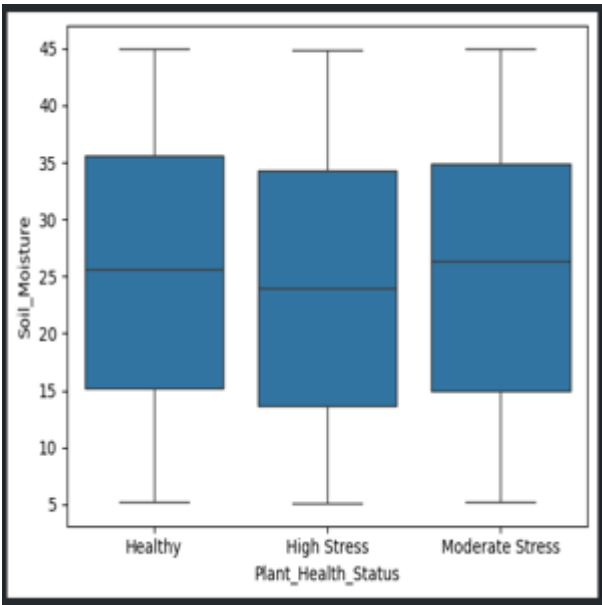


Figure 7. Agriculture Dataset Correlation

8. Dataset 3: Plant Health Biosensor Dataset

8.1. Findings

- The dataset shows a moderate predictive signal.
- It appears more aligned with sensor-driven use cases and may transfer better to constrained (TinyML-style) deployment than the weaker datasets.
- Requires the same quality checks as the others, but initial behavior is promising.

8.2. Key Metrics

- Baseline test accuracy (notebook run): 0.6295

8.3. Proof Images

```
Dataset Validation Summary
-----
Rows used: 1253
Features used: ['Soil_Moisture', 'light_absorption_ratio', 'Ambient_Temperature']
Target classes: [0, 1, 2, 3]
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.6295
```

Figure 8. Plant health BioSensor Summary

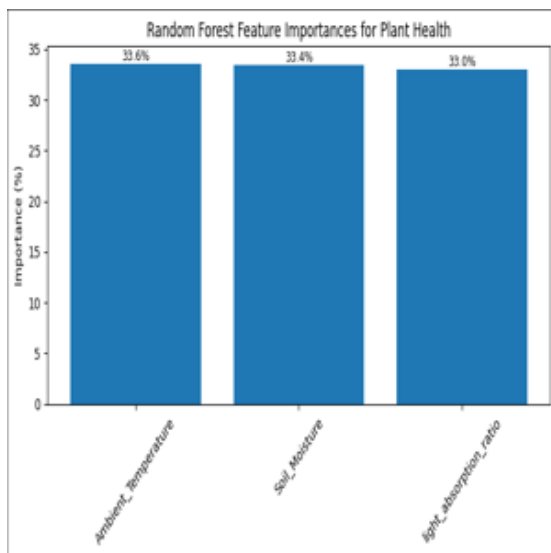


Figure 9. Plant health BioSensor Feature Importance

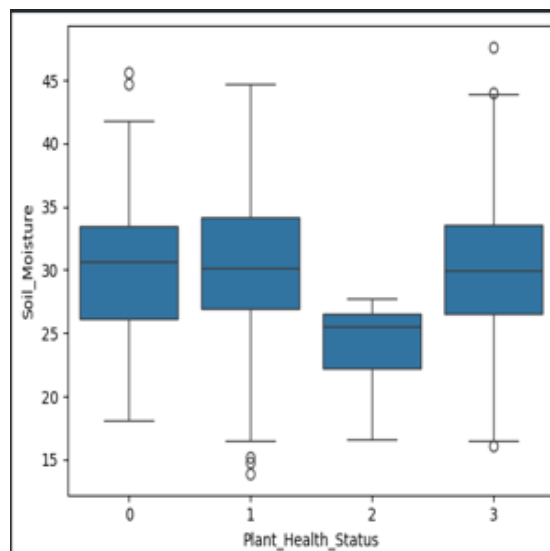


Figure 10. Plant health BioSensor Correlation

9. Dataset 4: Plant Health Data

9.1. Findings

- The dataset is one of the most usable options for the project.
- Raw features show imbalance in predictive power (single-feature dominance).
- After feature engineering, model behavior improves significantly and becomes more suitable for compact models.
- The engineered data set shows a more divided feature importance, rather than being mainly reliable upon one feature as the non-engineered model is.

9.2. Key Metrics

- Random Forest test accuracy: 0.8240
- TinyMLP test accuracy: 0.6958
- TinyRegression: MSE 0.2840, MAE 0.4148, R^2 0.5388

9.3. Engineered Version

As the given dataset appeared to be the most promising, it was decided to do some small data engineering. The type of data engineering that is used, feature engineering, allows a model to learn better patterns. Besides that, as the dataset is overly focused on one feature, relationships between features are introduced, interaction features. This is implemented as in the real world, certain features are connected, but this can not be read from the data, as the model has no idea of the real world relations.

Therefore, the engineered version focuses on finding the relation between different raw values. Rather taking the raw data as is, it looks for how can it be used to improve the models feature importance.

9.4. Proof Images

```
Dataset Validation Summary
-----
Rows used: 1200
Features used: ['Light_Intensity', 'Soil_Moisture', 'Humidity', 'Ambient_Temperature']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.8458
```

Figure 11. Plant Health Summary

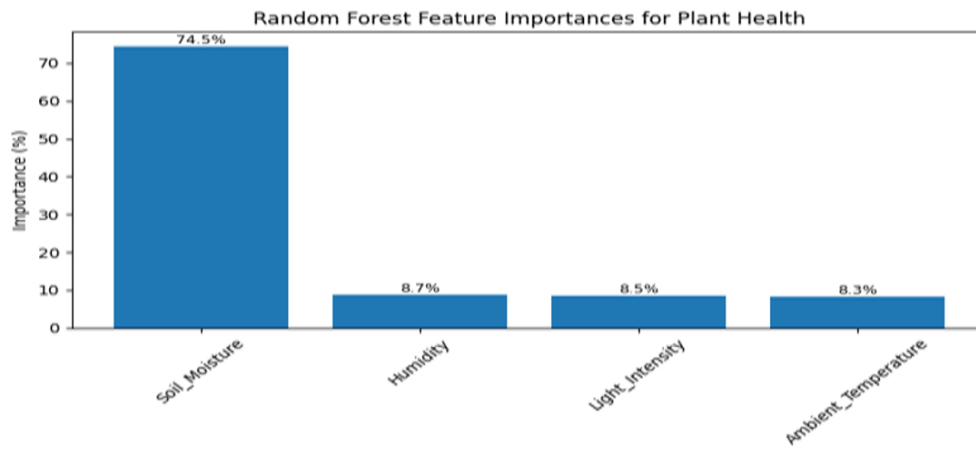


Figure 12. Plant Health Data Importance

Train set report:				
	precision	recall	f1-score	support
Healthy	1.00	1.00	1.00	239
High Stress	1.00	1.00	1.00	400
Moderate Stress	1.00	1.00	1.00	321
accuracy			1.00	960
macro avg	1.00	1.00	1.00	960
weighted avg	1.00	1.00	1.00	960
Test set report:				
	precision	recall	f1-score	support
Healthy	0.73	0.98	0.84	60
High Stress	0.99	0.73	0.84	100
Moderate Stress	0.84	0.89	0.86	80
accuracy			0.85	240
macro avg	0.85	0.87	0.85	240
weighted avg	0.87	0.85	0.85	240

Figure 13. Plant Health Data Report

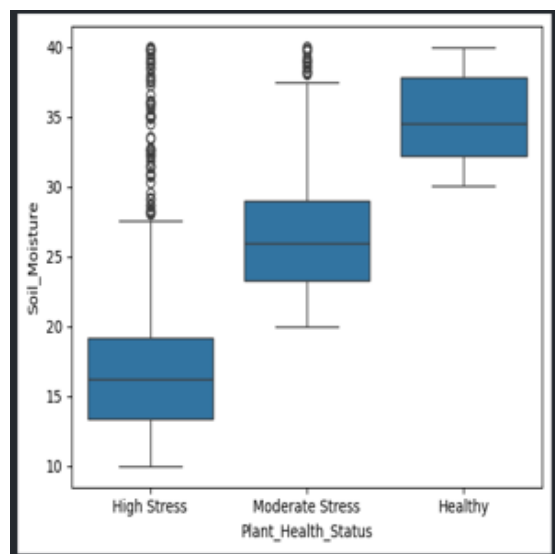


Figure 14. Plant Health Data Correlation

9.4.1. Engineered Version

```

Dataset Validation Summary
-----
Original rows: 1200
Rows used after cleaning: 1200
Features used: ['Soil_Moisture', 'humidity_stress', 'moisture_temp_interaction', 'light_moisture_interaction']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Train accuracy: 1.0000
Test accuracy: 0.8417

```

Figure 15. Plant Health Data Engineered Summary

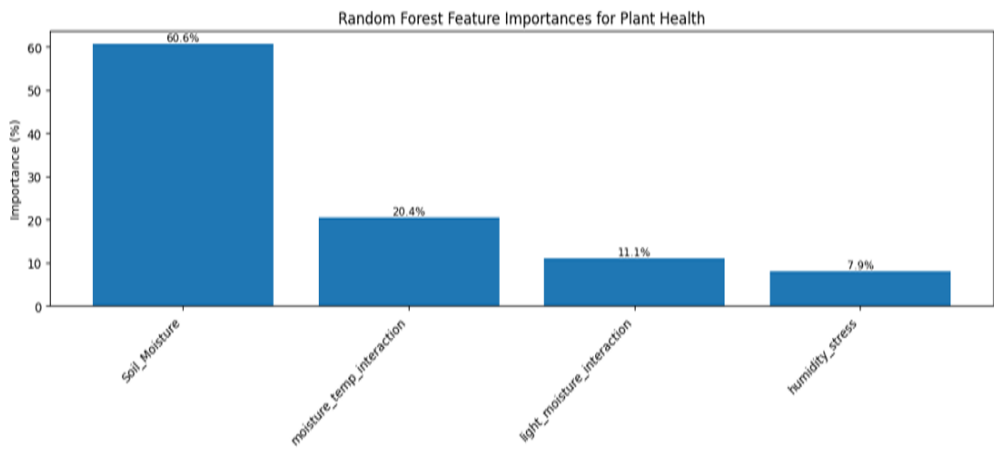


Figure 16. Plant Health Data Engineered Importance

	precision	recall	f1-score	support
Healthy	1.00	1.00	1.00	239
High Stress	1.00	1.00	1.00	400
Moderate Stress	1.00	1.00	1.00	321
accuracy			1.00	960
macro avg	1.00	1.00	1.00	960
weighted avg	1.00	1.00	1.00	960
Test set report:				
	precision	recall	f1-score	support
Healthy	0.73	0.98	0.84	60
High Stress	0.99	0.73	0.84	100
Moderate Stress	0.82	0.88	0.85	80
...				
accuracy			0.84	240
macro avg	0.85	0.86	0.84	240
weighted avg	0.87	0.84	0.84	240

Figure 17. Plant Health Data Engineered Report

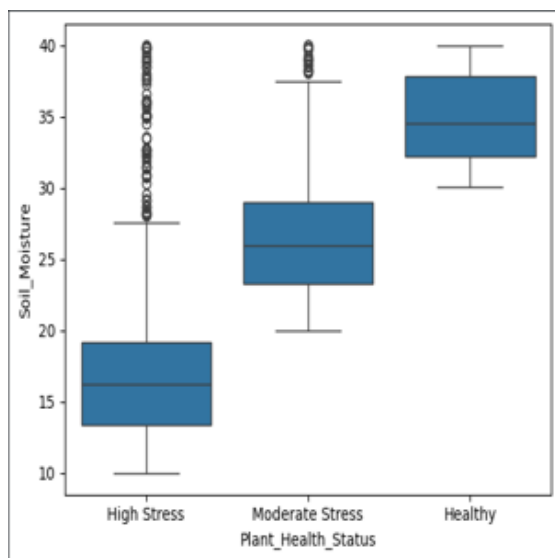


Figure 18. Plant Health Data Engineered Correlation

10. Cross-Dataset Comparison (TinyML / TinyRegression / Anomaly Detection)

The combined comparison supports the final dataset ranking by showing how each dataset behaves under:

- baseline classification (**RandomForest**)
- compact neural classification (**TinyMLP**)
- compact regression (**TinyRegression**)
- anomaly detection (**IsolationForest**)

model	accuracy	mse	mae	r2	anomaly_count	dataset
RandomForest	0.825000	NaN	NaN	NaN	NaN	Plant_Health
TinyMLP	0.695833	NaN	NaN	NaN	NaN	Plant_Health
TinyRegression	NaN	0.285050	0.414802	0.538795	NaN	Plant_Health
AnomalyDetection	NaN	NaN	NaN	NaN	701.0	Plant_Health
RandomForest	0.215000	NaN	NaN	NaN	NaN	Indoor_Plant
TinyMLP	0.205000	NaN	NaN	NaN	NaN	Indoor_Plant
AnomalyDetection	NaN	NaN	NaN	NaN	132.0	Indoor_Plant
RandomForest	0.312500	NaN	NaN	NaN	NaN	Agriculture
TinyMLP	0.365000	NaN	NaN	NaN	NaN	Agriculture
TinyRegression	NaN	0.700069	0.715652	-0.072995	NaN	Agriculture
AnomalyDetection	NaN	NaN	NaN	NaN	218.0	Agriculture

Figure 19. Datasets Comparison

The comparison image is used as supporting evidence that Plant Health Data (especially engineered) is the strongest candidate for lightweight deployment scenarios.

11. Executive Summary

Table 1. Executive Summary

Dataset	Baseline Signal	TinyML Readiness	Overall Verdict
Plant Health Data	Strong	Good	Recommended after feature engineering.
Indoor Plant Health & Growth Factors	Weak	Poor	Not recommended in current form (weak relationship to target).
Agriculture Dataset with Target	Weak-to-moderate	Limited	Usable for experiments, not strong for reliable prediction.
Plant Health Biosensor Dataset	Moderate	Promising	Potentially usable for constrained embedded models.

12. Final Recommendation

For the project objective (small, practical, and deployable predictive models), the Plant Health Data (engineered version) should be used as the primary dataset. It has the most potential, as from the original dataset validation, it showed promising results.

As back-up or exploratory, the Plant Health Biosensor data should be kept in mind. While it is not the best suitable for the given objective, it does show some potential, which with further engineer could be brought to light.

13. References

- [ml_google] Google. **ML Crash Course**. Available at: <https://developers.google.com/machine-learning/crash-course>
- [scikit_model] Scikit. **Model Selection**. Available at: https://scikit-learn.org/stable/model_selection.html
- [ibm_evaluation] IBM. **Model Evaluation**. Available at: <https://www.ibm.com/think/topics/model-evaluation>
- [kaggle_intro] Kaggle. **Intro To Machine learning**. Available at: <https://www.kaggle.com/learn/intro-to-machine-learning>
- [microcontroller_google] Google. **Microcontrollers Overview**. Available at: <https://ai.google.dev/edge/litert/microcontrollers/overview>
- [shap] Shap. **Shap**. Available at: <https://shap.readthedocs.io/en/latest/>
- [christophm_gh] ChristophM. **ML Interaction**. Available at: <https://christophm.github.io/interpretable-ml-book/interaction.html>
- [apxml_features] Apxml. **Feature Interaction**. Available at: <https://apxml.com/courses/intro-feature-engineering/chapter-5-feature-creation/interaction-features?>
- [metadesign_ml] MetaDesignSolutions. **Feature Engineering in ML**. Available at: <https://metadesignsolutions.com/feature-engineering-in-machine-learning>